



Smart Dam Water Level & Safety Monitoring (L&T Internship)

DAY 1 Milestone: **OVERVIEW**

☐ Milestone 1: System Architecture Design

Design the end-to-end IoT + AI system architecture for dam monitoring.

Define data flow between rainfall sensors, water-level sensors, pressure sensors, cloud, ML model, and dashboard.

Contribution Summary:

Member	Task	Completion%	Status
1.Dheenadayalan N	Documentation & Sensor Selection & Integration		
2.Jagath A	Circuit Design (Circuit Designer)		

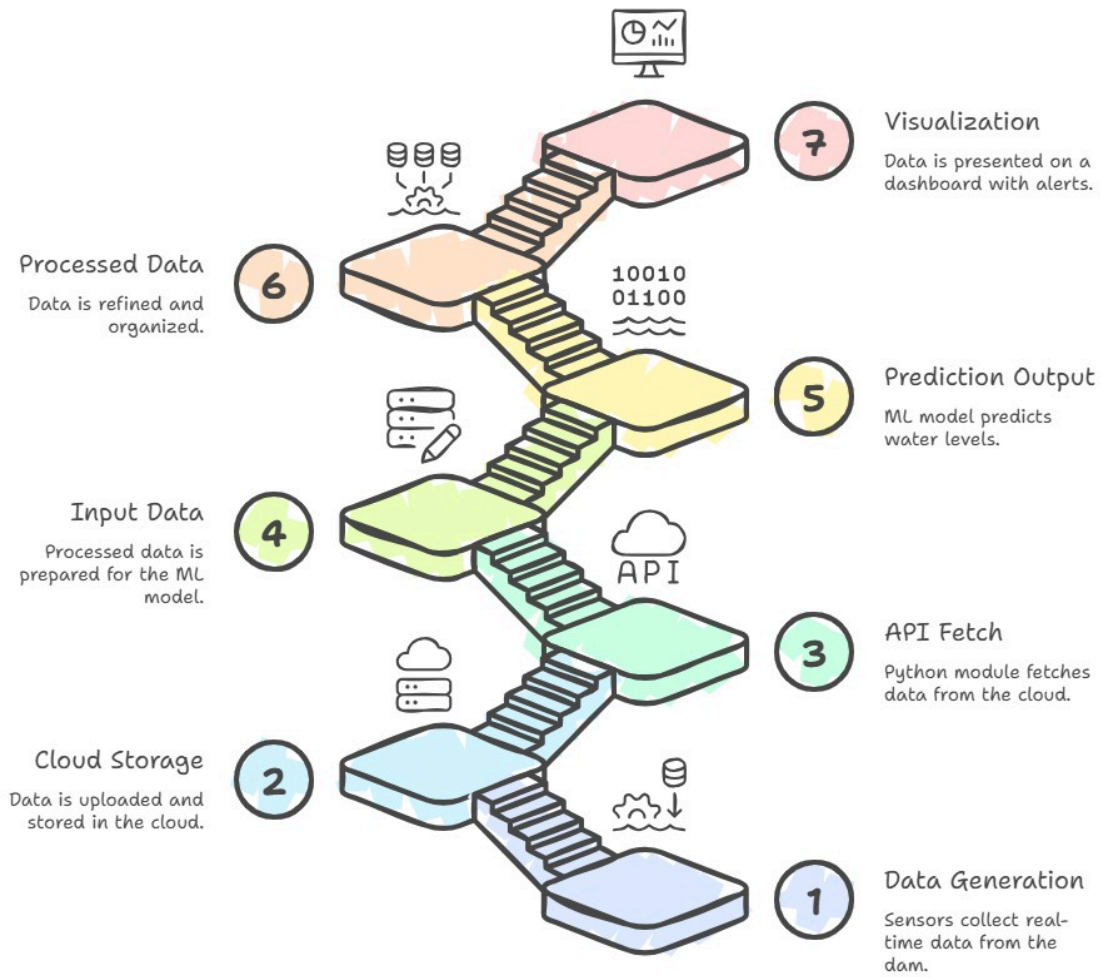
3.Kuncham Yaswanth Reddy	Data Flow &Progress Tracking		
4. Asam Deepak Reddy	System Architecture Design		

Components Table:

Layer	Component	Function
Sensor Layer	YL-83 Rain Sensor	Detect rainfall intensity
Sensor Layer	Water Level Sensor	Measure reservoir water level
Sensor Layer	Force Sensor (FSR)	Measure pressure/force
Edge Layer	ESP32-S3	Read sensors + process data + send to cloud
	WiFi (HTTP/MQTT)	Transmit data
Cloud Layer	ThingSpeak	Store and manage data
ML Layer	ML Model	Predict water level & risk
Application Layer	Power BI Dashboard	Visualization & alerts

Data Flow:

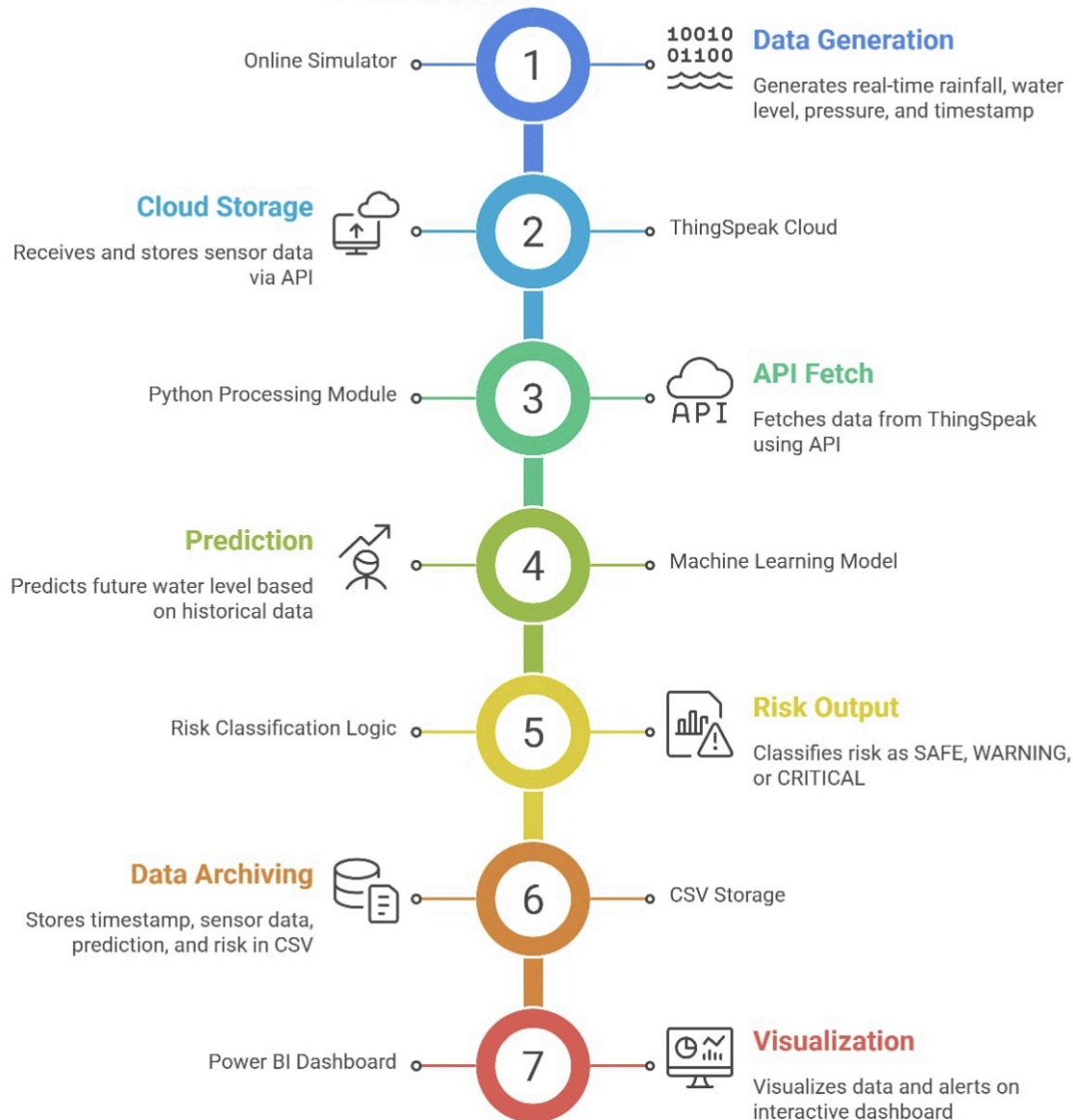
Smart Dam Monitoring System Data Flow



Made with Napkin

System Architecture:

Smart Dam Monitoring System- IoT + AI Architecture



Made with Napkin

Circuit Design using ESP32-S3 & Multi Sensor Integration:



Cirkit Designer IDE

Cirkit Designer IDE is an all-in-one circuit design tool that helps you create and share electronic circuits with ease.

 <https://app.cirkitdesigner.com/project/b9c5a56f-4149-4e9a-a77c-b200e91c1abf>

ThingSpeak Graph:

Smart Dam

Channel ID: 3328302
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